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Business Analytics

Experiment No.: 2

Title: Superstore Sales Analytics and Visualization Using Power BI

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Superstore Sales Analytics and Visualization Using Power BI

GitHub Repo: https://github.com/Amaraiah11/Business_Project_Analysis-

1. Introduction

Business data contains valuable information about sales, customers, products and profitability. Microsoft Power BI can be used to convert this data into interactive dashboards that help in understanding business performance and making better decisions.

In this experiment, the Superstore dataset was analysed using Power BI. The main objective was to study sales and profit performance through different dashboards and visualizations. The dashboards provide information about sales trends, profit, orders, quantity, customers, categories, sub-categories, regions and customer segments.

The Power BI report consists of three dashboards that provide different views of the business. The first dashboard focuses on overall sales performance, the second focuses on profitability and customer segments, and the third provides detailed geographical and product-level analysis.

2. Dashboard 1 – Superstore Sales Performance

The first dashboard provides an overall view of Superstore's business performance. It contains important Key Performance Indicators (KPIs) that give a quick summary of the business.

The main KPIs displayed in the dashboard include:

- Total Sales
- Total Profit
- Total Orders
- Total Quantity
- Average Order Value

These KPIs help in understanding the overall scale and performance of the business.

The dashboard also contains visualizations based on Category, Sub-Category, Product Name and Region. These visuals help compare the contribution of different product groups and geographical areas.

A line chart is used to display the sales trend over Order Date. This helps identify changes in sales performance over different time periods. Category and sub-category visualizations help determine which product groups contribute significantly to overall sales.

The regional analysis provides a comparison of business performance across different regions. Product-level analysis can also be used to identify products that contribute more to sales.

Purpose of the Dashboard

The main purpose of this dashboard is to provide a quick overview of business performance. Management can use it to monitor important KPIs and identify areas that require further investigation.

3. Dashboard 2 – Profitability and Segment Analysis

The second dashboard focuses mainly on profitability and customer segment analysis. It provides interactive filters that allow users to analyse the data according to different conditions.

The dashboard includes slicers for:

- Segment
- Order Date
- Region
- Sub-Category

These filters allow users to select specific segments, regions or time periods and analyse their performance.

A regional profit chart is used to compare Total Profit between regions. This helps identify regions that contribute more or less to overall profitability.

The dashboard also provides analysis based on Category and Segment. Customer segments can be compared to understand their contribution to sales and business performance.

Another important visualization is the scatter chart comparing Average Discount with Total Profit for different sub-categories. This analysis helps understand the relationship between discounting and profitability.

Higher discounts may not always result in higher profits. Therefore, analysing discount levels together with profit can help businesses develop better pricing and discount strategies.

Purpose of the Dashboard

The main purpose of this dashboard is to understand profitability and customer behaviour. It helps management analyse which regions, categories and customer segments are performing well and which areas may require improvement.

4. Dashboard 3 – Detailed Business and Geographical Analysis

The third dashboard provides a more detailed analysis of the Superstore business. It combines geographical, product and profitability information.

A geographical map is used to display information based on regions. The map provides information related to:

- Total Sales
- Total Profit
- Total Customers

This allows users to understand how business performance is distributed geographically.

The dashboard also contains a detailed matrix that provides information for Category and Sub-Category. The matrix includes measures such as:

- Profit Margin Percentage
- Total Sales
- Total Quantity
- Total Profit

This detailed view helps compare the performance of individual product categories and sub-categories.

A line chart is also included to analyse the sales trend over Order Date. This helps identify changes in sales over time.

A treemap is used to represent Sub-Categories based on Total Quantity. Larger sections represent sub-categories with a higher quantity contribution.

Purpose of the Dashboard

The purpose of this dashboard is to provide detailed information for business analysis. Users can identify high-performing and low-performing product groups and compare performance between different regions.

5. Findings, Conclusion and Recommendations

Findings

The Power BI dashboards provide a comprehensive view of Superstore's business performance. The analysis combines sales, profit, customer, product and geographical information.

The main findings from the dashboards include:

1. **Sales Performance:** Total Sales and sales trends provide an overview of the company's overall business activity.
2. **Profitability:** Total Profit and Profit Margin Percentage help evaluate whether sales are generating sufficient returns.
3. **Regional Performance:** Regional visuals help compare sales and profit across different geographical areas.
4. **Category Performance:** Category and Sub-Category analysis helps identify differences in product performance.
5. **Customer Segments:** Segment analysis helps understand how different customer groups contribute to business performance.
6. **Discount Analysis:** Comparing discounts with profit helps identify the possible impact of discounting on profitability.
7. **Time-Based Analysis:** Sales trends over Order Date help identify changes in business performance over time.

Recommendations

Based on the dashboard analysis, the following recommendations can be made:

- Regularly monitor Total Sales, Total Profit and Average Order Value.
- Analyse regions with comparatively lower profitability and identify the reasons for weaker performance.
- Review discount strategies to ensure that increased discounts do not negatively affect profit.
- Monitor the performance of individual categories and sub-categories.
- Use customer segment analysis to understand different customer groups and their contribution.
- Track sales trends regularly to identify changes in business performance.
- Use the detailed dashboard to support product-level and regional business decisions.

Conclusion:-

The Superstore Business Performance Analysis demonstrates how Microsoft Power BI can be used to transform business data into meaningful visual insights. The three dashboards provide different levels of analysis, ranging from an overall performance summary to detailed product and geographical analysis.

The use of KPI cards, charts, slicers, maps, matrices and treemaps makes the data easier to understand and analyse. The dashboards can help management monitor performance, identify problem areas and make data-driven decisions.

Overall, Power BI provides an effective and interactive platform for analysing Superstore business performance and presenting the results in a clear and understandable manner.

Dashboards:-

